CHAPTER – VIII

STRINGBUFFER AND STRINGBUILDER:

- StringBuffer and StringBuilder are used to create and modify String efficiently.
- String is immutable.

STRINGBUFFER – Multi – thread environment.

STRINGBUILDER – Single – thread environment(most cases).

STRINGBUFFER:

- StringBuffer is a predefined class in java used to create and mutable(modifiable) strings.
- It is also thread – safe, because all its methods are synchronized.

EXAMPLE:

```
public class StringBufferExample {
   public static void main(String[] args) {
      StringBuffer sb = new StringBuffer("Hello");
      sb.append(" World");
      System.out.println("Append: " + sb);
      sb.insert(5, " Java");
      System.out.println("Insert: " + sb);
      sb.replace(6, 10, "C++");
      System.out.println("Replace: " + sb);
      sb.delete(6, 10);
```

```
        System.out.println("Delete: " + sb);
        sb.reverse();
        System.out.println("Reverse: " + sb);
    }
}
```

STRINGBUILDER:

- StringBuilder is used to create mutable strings.
- It is not synchronized.

EXAMPLE:

```
public class StringBuilderExample {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello");
        sb.append(" World");
        System.out.println("Append: " + sb);
        sb.insert(5, " Java");
        System.out.println("Insert: " + sb);
        sb.replace(6, 10, "C++");
        System.out.println("Replace: " + sb);
        sb.delete(6, 10);
        System.out.println("Delete: " + sb);
        sb.reverse();
        System.out.println("Reverse: " + sb);
    }
```

}

THREAD:

- Thread is the smallest unit of program that can run independently.
- It allows multiple tasks at the same time.

EXAMPLE:

```
class MyThread extends Thread{
    public void show(){
        System.out.println(“Thread is running”);
    }
}
```

THREAD – SAFE:

- A code or object is thread – safe if multiple threads can use it at the same time without problems.

MULTITHREADING:

- Multithreading is the process of executing two or more threads at the same time within a single java program.

ADVANTAGES:

- Improves Performance
- Better CPU utilization
- Useful for real time Applications
- Allows background tasks

SYNCHRONIZATION:

- Synchronization means only one thread is allowed to use a method or block of code at a time.

EXAMPLE:

```
synchronized void print(){
    System.out.println(“Hello”);
}
```